

## 7. Integration Techniques

### 7.3. Trigonometric Substitutions

#### 7.3b Learning Objectives

- I can properly handle the bounds on a trigonometric substitution problem.

#### 7.3.4. Definite Integrals

We touched on it in our motivating example, but we have to be more careful with bounds on these substitutions. Of course, you can find the indefinite integral and deal with bounds separately if you are careful with notation. Here we will go through an example or 2 of a definite integral where we change the bounds. This saves a lot of writing, and means we don't have to rewrite the integral in terms of  $x$  at the end.

#### Example 7.3.1

Find

$$\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2 + 9)^{\frac{3}{2}}} dx$$

#### Solution

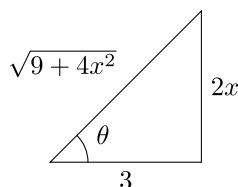
First we rewrite the integrand with something that has a root.

$$\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2 + 9)^{\frac{3}{2}}} dx = \int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(\sqrt{4x^2 + 9})^3} dx$$

Now we draw the triangle. Since the root has a plus sign (instead of minus) the root term goes on the hypotenuse of the triangle. That means the  $2x$  is either on the opposite or the adjacent side, and the trig function that avoids the root is tangent. So the substitution we need to use is

$$x = \frac{3}{2} \tan(\theta)$$
$$\frac{2x}{3} = \tan(\theta) = \frac{\text{opp}}{\text{adj}}$$

. Now we can tell the opposite side is  $2x$  and the adjacent side is 3. The triangle is



Ok, now to get the differential

$$x = \frac{3}{2} \tan(\theta)$$
$$dx = \frac{3}{2} \sec^2(\theta) d\theta$$

Lastly we need to rewrite the root expression as a trig function. The trigonometric function that includes the root term and the constant term in our triangle is cosine or secant. We know we already have secant in our problem, so we'll use that

$$\sec(\theta) = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{9+4x^2}}{3}$$

$$\Rightarrow 3\sec(\theta) = \sqrt{9+4x^2}$$

Now we have to actually make this substitution. Plugging in  $x$ ,  $dx$ , and  $\sqrt{9+4x^2} = 3\sec(\theta)$  gives us

$$\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(\sqrt{4x^2+9})^3} dx = \int_{\square}^{\square} \frac{\left(\frac{3}{2}\tan(\theta)\right)^3}{(3\sec(\theta))^3} \left(\frac{3}{2}\sec^2(\theta)\right) d\theta.$$

And the new bounds are found from

$$0 = \frac{3}{2}\tan(\theta) \Rightarrow \theta = 0$$

$$\frac{3\sqrt{3}}{2} = \frac{3}{2}\tan(\theta) \Rightarrow \sqrt{3} = \tan(\theta) \Rightarrow \theta = \frac{\pi}{3}$$

So now we need to solve the following integral

$$\begin{aligned} \int_0^{\frac{\pi}{3}} \frac{\left(\frac{3}{2}\tan(\theta)\right)^3}{(3\sec(\theta))^3} \left(\frac{3}{2}\sec^2(\theta)\right) d\theta &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\tan^3(\theta)}{\sec(\theta)} d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\sin^3(\theta)\cos(\theta)}{\cos^3(\theta)} d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \tan^2(\theta)\sin(\theta) d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} (\sec^2(\theta) - 1)\sin(\theta) d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\sin(\theta)}{\cos(\theta)} \sec(\theta) - \sin(\theta) d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \tan(\theta)\sec(\theta) d\theta - \frac{3}{16} \int_0^{\frac{\pi}{3}} \sin(\theta) d\theta \\ &= \frac{3}{16} [\sec(\theta)]_0^{\frac{\pi}{3}} - \frac{3}{16} [-\cos(\theta)]_0^{\frac{\pi}{3}} \\ &= \frac{3}{16} [2 - 1] + \frac{3}{16} \left[\frac{1}{2} - 1\right] \\ &= \frac{3}{32} \end{aligned}$$

### Example 7.3.2

Evaluate the following definite integral. Include a drawing of the triangle used in the trig sub.

$$\int_0^1 \sqrt{x - x^2} \, dx$$

#### Solution

We can use trig subs to help us solve even more complicated integrals! We've seen we need something with  $x$  squared and something without  $x$ .

Recall how to complete the square.

$$\begin{aligned} x - x^2 &= b - (x - c)^2 \\ &= b - (x - 2xc + c^2) \\ &= (b - c^2) + 2cx - x^2 \end{aligned}$$

So

$$2c = 1 \quad \text{and} \quad b - c^2 = 0$$

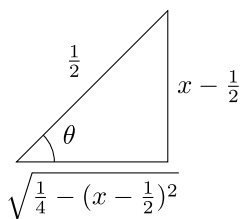
then

$$c = \frac{1}{2} \quad \text{and} \quad b = \frac{1}{4}$$

So we can rewrite

$$\int_0^1 \sqrt{x - x^2} \, dx = \int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} \, dx$$

Setting up the triangle, the hypotenuse is  $\frac{1}{4}$ . We know the adjacent side to be the square root term, and the opposite side to be  $x - \frac{1}{2}$ . Then the triangle is



The substitution we use is  $\frac{1}{2} \sin(\theta) + \frac{1}{2} = x$  so  $dx = \frac{1}{2} \cos(\theta) \, d\theta$ , and  $\sec(\theta) = \frac{\sqrt{\frac{1}{4} - (x - \frac{1}{2})^2}}{\frac{1}{2}}$ . The integrand becomes

$$\int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} \, dx = \int_{\square}^{\square} \frac{1}{2} \sec(\theta) \frac{1}{2} \cos(\theta) \, d\theta$$

The bounds become

$$0 = \frac{1}{2} \sin(\theta) + \frac{1}{2}$$

$$\sin(\theta) = -1$$

$$\theta = -\frac{\pi}{2}$$

and

$$1 = \frac{1}{2} \sin(\theta) + \frac{1}{2}$$

$$\sin(\theta) = 1$$

$$\theta = \frac{\pi}{2}$$

So the integral is an easy one it is

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{4} d\theta = \frac{\pi}{4}$$

### Tip

#### Procedure for Trigonometric Substitution

1. Draw a right triangle with appropriate side lengths and angle  $\theta$  labeled. Write down  $\sin(\theta)$ ,  $\cos(\theta)$ , and  $\tan(\theta)$ , in terms of the side lengths.
2. Write down the substitution for the variable  $x$ , calculate the differential  $dx$ , and specify the selected values of  $\theta$  for substitution.
3. Substitute the trigonometric expression and the calculated differential into the integrand. Simplify the results algebraically.
4. Integrate the trigonometric integral, keeping in mind the restrictions on the angle  $\theta$  for reversibility.
5. Use the right triangle to reverse the substitution in the integration result and convert it back to the original variable  $x$ . In particular, compositions of trig and inverse trig functions of  $x$  should be avoided (final answers should be in fully simplified form).

#### 7.3b Section Summary:

- We did a definite integral that needed a trig substitution.
- We used complete the square to evaluate a more complicated integral with a square root.